Date: __/__/__

Assignment - 4

Unit - 4

Q1 What is ~~fast~~ setting of cement?

Ans When mixed with water, cement sets to a hard mass. It first forms a plastic mass which hardness after some time due to 3-dimensional cross-links between the --Si-O-Si- & --Si-O-Al-- chains. The first setting occurs within 24 hours whereas the subsequent hardening requires a fortnight when it covered by a layer of water. This transition from plastic to solid state is called setting.

Reaction involved in setting of cement.

(i) $3CaO \cdot Al_2O_3 + 6H_2O \longrightarrow 3CaO \cdot Al_2O_3 \cdot 6H_2O$

(ii) $2CaO \cdot SiO_2 + xH_2O \longrightarrow 2CaO \cdot SiO_2 \cdot xH_2O$

(iii) $3CaO \cdot SiO_2 + H_2O \longrightarrow Ca(OH)_2 + 2CaO \cdot SiO_2$

(iv) $3CaO \cdot Al_2O_3 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2Al(OH)_3$

(v) $3CaO \cdot Al_2O_3 + 3CaSO_4 + 2H_2O \longrightarrow 3CaO \cdot Al_2O_3 \cdot 3CaSO_4 \cdot 2H_2O$

The fast-setting tricalcium aluminate is removed to slow down the setting process. A quick setting will give rise to crystalline hydrated calcium aluminate. A slower setting yields the colloidal gel that imparts water strength to the set mass. Thus gypsum helps in ~~equal~~ regulating the setting time of cement.

Date : __/__/__

Q2 What should be the flash point of good lubricant

Ans The flash point of a volatile material is the lowest temperature at which vapours of the material will ignite for a moment when an ignition source brought near to it. The lubricating oil should have flash point reasonably above its warking temp. It should be high.

Q3 Write chemical composition of constituents of cement.

Ans

Constituent	Range by mass	Function
i) ~~lime~~ Lime	60-70	↑ strength
ii) Silica (SiO_2)	17-25	↑ strength & prolong setting
iii) Alumina (Al_2O_3)	3-8	↑ strength but reduces setting time.
iv) Iron-oxide (Fe_2O_3)	2-4	Impart greay colour
v) MgO	1-5	↑ strength & hardness contribute to soundness
vi) SO_3 (Sulphuric oxide)	1-3	Imparts soundness if present in small amt.

Date : __/__/____

(viii) Alkali oxide 0.3–1.5 Causes eff.
Na_2O / K_2O

Q4 What is glass wool & write its uses?

Ans Fibrous wool like material composed of mixturing — led fine filament of glass. These are made by action of steam jet on creeping molten glass down from very fine holes.

Properties

(a) Non combustible & fire proof.
(b) Poor conductor of electricity.
(c) Heat proof.
(d) Resistance to most chemicals.
(e) Very high tensile strength.

Uses :- Electrical & sound insulation, heat insulation, in domestic & industrial appliances, for filteration of corrosive liquids like acids. To manufacture of glass fibre.

Q5 Write chemistry of setting & hardening of cement.

Ans Setting :- When cement is mixed with water & allow to stand, it sets into a hard & rigid mass due to chemical reaction & gel formation.

Hardening :- Converting into compact rock like material & developing of strength due to crystallisation is known as hardening.

Date : __/__/__

The process of setting & hardening of cement are believed to be partly a chemical & part, a physical changes.

Chemical change - Hydrolysis
Physical change - Formation

Hydrolysis -
$$3CaO \cdot SiO_2 + 3H_2O \rightarrow 3CaO \cdot 2SiO_2 + Ca(OH)_2$$

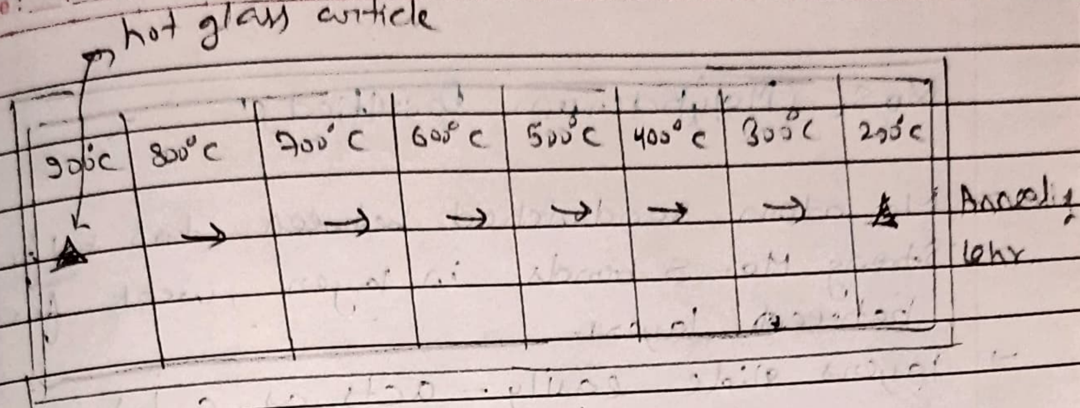
Hydration :-
$$3CaO \cdot Al_2O_3 + 6H_2O \rightarrow 3CaO \cdot Al_2O_3 \cdot 6H_2O$$
(crystal)

Q.6 Write a short note on annealing of glass.

Ans Annealing is a crucial heat treatment process used in glass manufacturing to relive internal stresses that develop during the cooling & shaping of glass. When hot glass is cooled too quickly, uneven temp. distribution can cause stress, making the glass brittle & prone to cracking or breaking.

In the annealing process, the glass is slowly cooled in a controlled environment, typically in an annealing oven or lehr. The temp is maintained just below the glass's softening point and gradually reduced over time. This allows the internal stresses to relax & ensures uniform temp distribution throughout the material.

Date: / /



Q7 Explain extreme-pressure lubrication?

Ans When the machine runs under the conditions of high loads the temp. rises between the moving surface of the metals. Thus, under such severe conditions ordinary fluid film or boundary fluid film is not effective because it may vapourize or fail to stick on the metal surfaces

Properties of Lubricants :-

1. Viscocity
2. Ash content
3. Acid value
4. Carbon residence
5. Oiliness
6. Velotility
7. Saponification
8. Flash & fire point
9. Aniline point
10. Cloud & pour point

Q8 Write structure of graphite & MOS_2, why are these used as solid lubricants.

Ans Graphite & molybdenum disulphate (MOS_2)

Graphite :- made of carbon atoms in hexagonal layer

Strong ~~bong~~ bonds within layers, weak forces between layer

layers slide easily - reduces friction

Date: / /

MoS_2 (Molybdenum Disulfied):-

- Mo atom sandwiched between two sulfur layer
- Strong Mo-S bonds in layer, weak forces between layers.
- layers slide easily - acts as a lubricant.

Use - layered structure allows easy sliding reduce friction & wear.
Work in dry, high-temp or vaccum condition.

Q9 Describe manufacturing of cement by Rotary kiln Process.

Ans Raw material - limestone ($CaCO_3$), clay (Alumina, silica) and other additives.
crushed & mixed in correct properties.

Raw material Preparation:-
Grinding:- The mixture is ground into a fine powder (called raw meal)

Rotary kiln Process:-
Raw meal is fed into a rotary kiln (a long, rotating cylinder slightly inclined)
Three main zone.
(i) Drying zone - Removes moisture from raw meal using hot gases.

2. Heating (Calcination) Zone :-
Temperature - 800-1000°C
Limestone decomposes -

$$CaCO_3 \rightarrow CaO + CO_2$$

iii. Burning (Clinkering) Zone
Temperature - 1400°C - 1500°C
material react to form clinker (nodules of calcium silicates etc.)

4. Cooling Clinker is cooled using air.
Sudden cooling is known as annealing.

5. Grinding :- Clinkers is ground with gypsum to from cement.

6. Packaging :- Final cement is packed & transported